Resolved Principal Mass Before Width Grouping:
Exact Endpoint Return and an Inverse-Length Equivalence

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Abstract

The positive multiplicative-resonance route contains the unresolved principal mass

$$\mathfrak{P}_X = \sum_{\beta} \frac{2H_{q_X}(N_\beta)}{q_X - 1} \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}, \qquad H_q(N) = \min\left\{\frac{q-1}{2}, \left\lfloor \frac{q}{N} \right\rfloor\right\}.$$

We open this quantity before grouping equal widths or multipliers and obtain an exact endpoint/long-fiber disintegration.

For $N = 1, 2$, one has

$$\frac{2H_q(N)}{q-1} = 1, \qquad g_q(H_q(N)) = 0.$$

Thus every remaining invertible atom on these fibers is purely principal and enters with its full physical weight. For $3 \le N \le q$, the elementary but sharp uniform comparison

$$\frac{1}{N} \le \frac{2\lfloor q/N \rfloor}{q-1} \le \frac{2q}{(q-1)N}$$

holds. If

$$\mathfrak{E}_X = \sum_{\substack{\beta : N_\beta \le 2 \\ z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}, \qquad \mathfrak{L}_X = \sum_{\beta : 3 \le N_\beta \le q_X} \frac{1}{N_\beta} \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z},$$

then

$$\boxed{\mathfrak{E}_X + \mathfrak{L}_X \le \mathfrak{P}_X \le \mathfrak{E}_X + \frac{2q_X}{q_X - 1}\mathfrak{L}_X.}$$

Consequently the principal gate is equivalent, up to absolute constants and in the same physical normalization, to controlling the actual endpoint mass and inverse-length mass separately.

This is an exact L1 reduction, not a proof of either bound. We also give sharp positive countermodels showing that the $1/N_\beta$ factor alone cannot control a polynomial number of branches, and that a long-fiber estimate cannot return the $N = 1, 2$ mass. No L2 fixed-shift estimate or parity breakthrough is claimed.

Keywords: principal resonance; resolved length; endpoint fiber; inverse-length mass; positive majorant; fixed shift.

Literal-frame convention. Every sum uses the actual $q_X \nmid u$ opened atoms, their physical weights, one prescribed $h_0 \neq 0$, and one inherited global normalization. No ambient branch count replaces an opened mass, and nothing is specialized to $h_0 = 2$.

1 The unresolved principal term

TPC-99 represents nonconstant resonance by the incidence window

$$E_H = \{\pm 1, \dots, \pm H\} \subset \mathbb{F}_q^\times.$$

TPC-100 then opens each frequency-free branch $\beta = (\theta, c, \kappa)$ into atoms

$$p = (\beta, z), \qquad w_{\beta,z} \geq 0, \qquad w_\beta = \sum_{z \in I_\beta^*} w_{\beta,z},$$

without changing its resolved length

$$N_\beta = \# I_\beta^*$$

or its resonance width $H_q(N_\beta)$ [1, 5]. After separately returning the $q_X \mid u_{\beta,z}$ submass, the remaining principal ledger is

$$\mathfrak{P}_X = \sum_{\substack{\beta \\ 1 \leq N_\beta \leq q_X}} \rho_{q_X}(N_\beta) \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}, \qquad \rho_q(N) := \frac{2H_q(N)}{q-1}. \tag{1}$$

The frequency-filter factor $B_X = \|b_X\|_1 \ll 1$ is not included in $\mathfrak{P}_X$.

TPC-102 proposes opening (1) before grouping by equal width. That order matters. Grouping first hides the fact that full-width resonance occurs at $N = 1, 2$ and that the remaining coefficient is exactly comparable to $1/N$ [2, 3].

2 The exact width coefficient

Let q be an odd prime and $1 \leq N \leq q$. Recall

$$H_q(N) = \min\left\{\frac{q-1}{2}, \left\lfloor \frac{q}{N} \right\rfloor\right\}.$$

Lemma 2.1 (The $N = 1, 2$ full-width endpoint lengths)**.** *For $N = 1, 2$,*

$$H_q(N) = \frac{q-1}{2}, \qquad \rho_q(N) = 1.$$

For $3 \leq N \leq q$,

$$H_q(N) = \left\lfloor \frac{q}{N} \right\rfloor.$$

For $q \geq 5$, these are the only full-width lengths. At the degenerate prime $q = 3$, $N = 3$ also has $H_3(3) = (q-1)/2$; it is nevertheless included correctly in the $N \geq 3$ inverse-length ledger below.

Proof. For $N = 1$, the minimum in $H_q(N)$ is visibly $(q-1)/2$. Since q is odd, $\lfloor q/2 \rfloor = (q-1)/2$, proving the same assertion for $N = 2$. If $N \geq 3$, then

$$\left\lfloor \frac{q}{N} \right\rfloor \leq \left\lfloor \frac{q}{3} \right\rfloor \leq \frac{q-1}{2}.$$

$\square$

Lemma 2.2 (Sharp inverse-length comparison)**.** *For every odd prime q and $3 \leq N \leq q$,*

$$\frac{1}{N} \leq \rho_q(N) \leq \frac{2q}{(q-1)N} \leq \frac{3}{N}. \tag{2}$$

The first inequality also holds for $N = 1, 2$.

Proof. Write $k = \lfloor q/N \rfloor$ and $q = kN + r$ with $0 \le r < N$. Then

$$2Nk - (q-1) = kN - r + 1 \ge N - r + 1 > 0.$$

Thus

$$\rho_q(N) = \frac{2k}{q-1} \ge \frac{1}{N}.$$

The upper bound follows from $k \le q/N$. Finally $2q/(q-1) \le 3$ for $q \ge 3$. At $N = 1, 2$, use lemma 2.1. ☐

Remark 2.3 (Optimal constants)*.* The lower constant 1 cannot be replaced uniformly by a number larger than 1: for $N = (q+1)/2$,

$$N\rho_q(N) = \frac{q+1}{q-1} \longrightarrow 1.$$

The asymptotic upper constant 2 is also sharp, for example at $N = q$, where $N\rho_q(N) = 2q/(q-1)$.

3 Resolved disintegration

For one branch define its actual remaining invertible mass

$$M_\beta^{\mathrm{inv}} = \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}.$$

This may be smaller than the full branch mass w_β; no replacement by N_β or an ambient count is made.

Definition 3.1 (Endpoint and inverse-length ledgers)**.** Set

$$\mathfrak{E}_X = \sum_{\substack{\beta \\ N_\beta = 1,2}} M_\beta^{\mathrm{inv}}, \tag{3}$$

$$\mathfrak{L}_X = \sum_{\substack{\beta \\ 3 \le N_\beta \le q_X}} \frac{M_\beta^{\mathrm{inv}}}{N_\beta}. \tag{4}$$

We also write

$$\mathfrak{E}_X = \mathfrak{E}_{1,X} + \mathfrak{E}_{2,X}$$

for the exact $N = 1$ and $N = 2$ subledgers.

Theorem 3.2 (Exact endpoint/long-fiber disintegration)**.** *The actual principal mass satisfies the identity*

$$\mathfrak{P}_X = \mathfrak{E}_{1,X} + \mathfrak{E}_{2,X} + \sum_{\substack{\beta \\ 3 \le N_\beta \le q_X}} \frac{2\lfloor q_X/N_\beta \rfloor}{q_X - 1} M_\beta^{\mathrm{inv}}. \tag{5}$$

Consequently,

$$\boxed{\mathfrak{E}_X + \mathfrak{L}_X \le \mathfrak{P}_X \le \mathfrak{E}_X + \frac{2q_X}{q_X - 1}\mathfrak{L}_X \le \mathfrak{E}_X + 3\mathfrak{L}_X.} \tag{6}$$

Proof. Insert lemma 2.1 into (1) for $N_\beta = 1, 2$, and insert its long-fiber formula for $N_\beta \ge 3$. This gives (5). Apply lemma 2.2 term by term to the nonnegative literal weights. ☐

Corollary 3.3 (Equivalent formulation of Gate P)**.** *The following are equivalent:*

$$\mathfrak{P}_X \le X^{o(1)} Q_X^2,$$

$$\mathfrak{E}_X + \mathfrak{L}_X \le X^{o(1)} Q_X^2,$$

$$\mathfrak{E}_X \le X^{o(1)} Q_X^2 \quad \text{and} \quad \mathfrak{L}_X \le X^{o(1)} Q_X^2.$$

All three statements have the same quantifier scope and physical normalization.

Proof. Use (6), nonnegativity, and the fact that fixed constants are absorbed in $X^{o(1)}$. $\square$

Remark 3.4 (What "equivalent" does not mean). Corollary 3.3 is an exact reduction of one sufficient positive gate. It does not identify that gate with the original signed low-window sum. A failure of $\mathfrak{E}_X$ or $\mathfrak{L}_X$ may stop the termwise positive route while signed frequency or branch cancellation remains possible.

4 The endpoint mass is exactly principal

TPC-101 defines

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}}.$$

Proposition 4.1 (Exact $N = 1,2$ return)**.** *For $N_\beta = 1,2$,*

$$g_{q_X}(H_{q_X}(N_\beta)) = 0.$$

Moreover, for the fixed positive frequency weight b_X, the endpoint contribution to the multiplicative incidence carrier is exactly

$$B_X\mathfrak{E}_X, \qquad B_X = \sum_{r\in\mathbb{F}_{q_X}^\times} b_X(r). \tag{7}$$

Proof. By lemma 2.1, $H = (q_X - 1)/2$, so the displayed formula for g_q vanishes. At this width,

$$E_H = \mathbb{F}_{q_X}^\times.$$

Hence the incidence indicator is one for every pair (r,ω), and its bilinear form is the product of the two total masses. The opened census mass at these two lengths is $\mathfrak{E}_X$. $\square$

Remark 4.2 (No centered theorem can pay the endpoint). The vanishing of $g_q(H)$ is not a saving on $\mathfrak{E}_X$. It says that the incidence operator has only its principal contribution at full width. Every surviving endpoint atom therefore has to be returned through (7), with its full physical weight.

5 Consequences of the atom cap

TPC-103 proves on the same actual support that

$$W_X := \max_{\beta,z} w_{\beta,z} = X^{o(1)}.$$

This is the literal pointwise result of TPC-103 [4]. Let

$$\mathcal{B}^{\mathrm{inv}}_{\le 2,X} = \{\beta : N_\beta \le 2,\ M_\beta^{\mathrm{inv}} > 0\}, \qquad \mathcal{B}^{\mathrm{inv}}_{\ge 3,X} = \{\beta : 3 \le N_\beta \le q_X,\ M_\beta^{\mathrm{inv}} > 0\}.$$

Proposition 5.1 (Exact-count sufficient conditions)**.** *One has*

$$\mathfrak{E}_X \le 2W_X|\mathcal{B}^{\mathrm{inv}}_{\le 2,X}|, \tag{8}$$

$$\mathfrak{L}_X \le W_X|\mathcal{B}^{\mathrm{inv}}_{\ge 3,X}|. \tag{9}$$

Thus a subpower atom cap plus an $X^{o(1)}Q_X^2$ bound for each actual branch census is sufficient for Gate P.

Proof. A branch of length N_β contains at most N_β opened atoms. Hence

$$M_\beta^{\mathrm{inv}} \le N_\beta W_X.$$

For $N_\beta \le 2$, sum this inequality directly. For $N_\beta \ge 3$, divide by N_β before summing. $\square$

Remark 5.2 (Why this is not an ambient-count replacement). Proposition 5.1 is only a sufficient corollary after the exact weighted identities have been proved. It does not authorize replacing M_β^{inv} by N_β in (5). A future proof should use the literal weights whenever they improve the crude branch census.

6 Sharp positive obstructions

Proposition 6.1 (Inverse length does not eliminate branch count)**.** *For every $M \geq 1$ and $3 \leq N \leq q$, there is an abstract nonnegative resolved census with M branches of length N, unit atom cap, and*

$$\mathfrak{L}_X = M, \qquad \mathfrak{P}_X = M\, N\rho_q(N) \asymp M.$$

Thus the $1/N$ factor alone gives no bound independent of the number of active branches.

Proof. Give every branch exactly N atoms of weight one. Then $M_\beta^{\mathrm{inv}} = N$, so its contribution to $\mathfrak{L}_X$ is one. The formula for $\mathfrak{P}_X$ follows from (1), and lemma 2.2 bounds $N\rho_q(N)$ between 1 and 3. □

Proposition 6.2 (Long fibers cannot return the endpoints)**.** *For every $M \geq 1$ there is an abstract nonnegative census with unit atom cap and M one-point branches such that*

$$\mathfrak{L}_X = 0, \qquad \mathfrak{E}_X = \mathfrak{P}_X = M.$$

Proof. Take M branches with $N_\beta = 1$, one atom of weight one on each, and apply lemma 2.1. □

Remark 6.3 (Scope of the countermodels)*.* The two constructions are L0 sharpness examples for the information used in a positive inequality. They do not assert that the actual TPC carrier contains such a family and are not stop certificates for the physical route. They prove that an actual mass or branch-incidence theorem is still needed.

7 Finite exact certificate

The script `experiments/tpc104_principal_mass_audit.py` checks, using exact rational arithmetic:

- every $1 \leq N \leq q$ for a deterministic list of odd primes;
- the endpoint identities and both inequalities in (2);
- the exact disintegration on deterministic weighted branch families; and
- the two countermodel formulas.

Its JSON output is a finite regression certificate, not numerical evidence that the actual growing ledgers satisfy their target.

8 Route decision and claim boundary

Established

- The $N = 1, 2$ invertible mass is returned exactly as a purely principal term.
- The long principal coefficient is uniformly and sharply comparable to $1/N_\beta$.
- Gate P is equivalent to the pair of actual weighted bounds $\mathfrak{E}_X, \mathfrak{L}_X \leq X^{o(1)} Q_X^2$.
- The TPC-103 cap reduces crude control of those ledgers to actual branch censuses, but does not prove the censuses.

These are L0 coefficient identities and an L1 lossless reduction on the actual fixed-h_0 carrier.

Not established

- Neither $\mathfrak{E}_X$ nor $\mathfrak{L}_X$ is bounded at the target scale for the growing physical packet.
- No polynomial lower bound for either actual ledger is proved, so the positive route is not declared stopped.
- No cross-map estimate, signed affine Möbius cancellation, full-block return, or strict endpoint closure is proved.
- No L2 fixed-power saving, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion follows.

The next exact task is to quotient identical affine maps without losing their input profile or source provenance. Only after that quotient is lossless should the genuinely distinct-map incidence be attacked, as prescribed by TPC-MVP1 [6].

References

[1] Liang Wang. Invertible Ultra-Divisor Linearization on the Nonconstant Branch: Exact Orbit Preimages, Diagonal Cell Energy, and the Cross-Cell Gate. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-100-invertible-ultra-multiplier-linearization. TPC-100.

[2] Liang Wang. Wiener-to-Hilbert–Schmidt Transfer for Multiplicative Resonance: Width-Sensitive Dispersion and the Long-Fiber Gate. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-101-wiener-hilbert-schmidt-resonance. TPC-101.

[3] Liang Wang. A Literal Resonance-Stage Audit for Fixed-Shift Prime Dynamics: The Three-Ledger Inequality, Exact Route Exclusions, and the Next Falsifiable Gates. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-102-literal-resonance-stage-audit. TPC-102.

[4] Liang Wang. The Normalized Opened-Atom Cap in the Literal Resonance Carrier: A Pointwise Factor Audit and Diagonal Transfer. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-103-normalized-opened-atom-cap. TPC-103.

[5] Liang Wang. Multiplicative Resonance Incidence at the No-Wrap Prime: Exact Character Spectrum, Parity Nullspace, and the Native Dispersion Gate. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-99-multiplicative-resonance-incidence. TPC-99.

[6] Liang Wang. TPC-MVP1: A Falsifiable Proof Architecture for Fixed-Shift Prime-Pair Correlations: Verified Interfaces, Explicit Working Hypotheses, and Stop–Go Tests at the Parity Frontier. *Prime Dynamics Theory*, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-mvp1-fixed-shift-proof-architecture. TPC-MVP1.